Morgan Smith

Splitting the Bill: A Signalling Model of Heterosexual Second-Date Decision-Making

Introduction, Motivation, and Theoretical Framework

When two parties agree to a first date, neither knows whether they will want a second or not. By the end of the first date, the woman has decided, but the man can not directly observe this assessment. His second-date decision therefore constitutes a choice under one-sided incomplete information. This paper models this interaction as a Beer-Quiche signalling game (Cho and Kreps, 1987) in which the woman is the informed sender and the man is the uninformed receiver.

The game has two players. Nature draws the woman's type $\theta \in \{\theta_I, \theta_U\}$ (Interested or Uninterested). The man proposes to split the bill; this proposal is exogenous and costless to him. The woman responds with signal $s \in \{A, R\}$, accepting or rejecting the split. The man observes s, updates his beliefs via Bayes' rule, and chooses action $a \in \{Y, N\}$: whether to propose a second date or not.

The bill-split proposal is not strategically neutral: it carries psychological costs and benefits that differ systematically across the two types. A woman who wishes to see him again (type I) faces the stigma of appearing mercenary if she rejects a split too readily. A woman who is not interested (type U) must weigh guilt aversion against the recuperative value of a complimentary meal. The guilt-aversion faced by the uninterested woman scales modestly with the male's attractiveness via the beauty premium found in economic allocation games, as does the loss aversion risk of appearing mercenary for the interested woman (Zahavi, 1975; Battigalli and Dufwenberg, 2007; Shang and Zhang, 2024). Conversely, acceptance of a split is relatively cheap for a type I woman, she loses the cost of her half of the bill, but trades this for the chance of future romantic connection.

These opposing forces produce distinct preference orderings over outcomes. The type I woman's general preference order is: (2nd date, full bill) ≻ (2nd date, split) ≻ (no 2nd date, full bill) ≻ (no 2nd date, split). She prefers a second date over not, but if there is no second date, she prefers to at least have her meal paid for. The type U woman's general preference order is: (no 2nd date, full bill) ≻ (2nd date, full bill) ≻ (no 2nd date, split) ≻ (2nd date, split). In an ideal case, she is not asked on a second date as she incurs a small "awkwardness" cost from having to reject the man, however this cost is lower than the gain of the free meal – meaning she prefers to reject the split always. However, as I mentioned above, the payoffs of the women vary with the attractiveness of the man, which means for certain P value thresholds, the ordinal ranking of the payoffs may be subject to change. This is empirically supported and intuitive: for an interested woman dating a very attractive male, the relative value of a free meal falls drastically in comparison to the risk of losing the chance of a second date by appearing mercenary. The beauty premium means even an uninterested woman feels more inclination to split the bill with a more attractive man – which I account for in the guilt aversion function (Battigalli & Dufwenberg, 2007; Shang & Zhang, 2024). Because these ordinal preferences generate sufficiently distinct acceptance rates across types, the bill-split response becomes a credible signal, and the man can rationally condition his second-date decision on what he observes, based on his self-believed attractiveness.

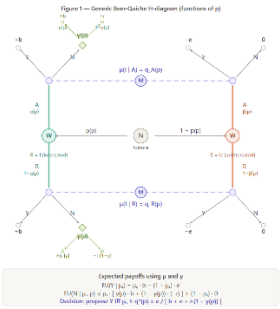
The man's decision problem is governed by four payoff states. When he proposes and the woman is type I, he receives a correct-match payoff of $+b = +12$. When he proposes and the woman is type U, he incurs an embarrassment cost of $-e = -20$. Social rejection hurts more than a match pleases, consistent with the disproportionate psychological weight of rejection documented by Baumeister et al. (2001). If the man does not propose and the woman is type U, the payoff is zero. If the man does not propose and the woman is type I, he incurs a missed-connection cost that is not a fixed penalty but an endogenous one: $-r\cdot(1 - \gamma(p))$, where the base regret parameter $r = 8$ is discounted by $\gamma(p)$, the probability that a genuinely interested woman self-initiates if the man stays silent. The effective regret cost therefore falls from 6.80 at minimum attractiveness to just 0.96 at maximum attractiveness, as more attractive men become increasingly likely to have a missed connection self-corrected by the woman. A third structural constraint, $r < e$, ensures that the fear of missing a connection never dominates the fear of an unwanted proposal; if it did, the man would propose almost indiscriminately and the bill signal would lose all screening power.

The Model, Decision Threshold and Equilibrium Structure

The prior probability that the woman is type I is $p(p)$, increasing in male attractiveness $p \in [0, 1]$: $p(0) = 0.60$, $p(0.5) = 0.80$. I anchor this to the empirical 20% first-date non-match rate at average attractiveness (0.5) (ZipDo, 2025), plateauing at 0.95 for $p \geq 0.875$. The type U acceptance rate $\beta(p) = 0.12 + 0.20p$ captures the tension between guilt aversion (a player's payoff falls below the counterparty's expectation), and recuperation (the free meal partially offsets the type U woman's sunk dating costs (Hochschild, 1983)).

The type I acceptance rate is: $\alpha(p) = \beta(p) + (1 - \beta(p)) \cdot \sigma(13p - 6.706)$, where $\sigma(\cdot)$ is the logistic function. The steepness parameter $k = 13$ is grounded in loss-aversion theory (Kahneman and Tversky, 1979): the fear of appearing mercenary is negligible at low p but rises sharply as the man gets more attractive, consistent with prospect-theoretic reference dependence. The intercept solves for the anchor $\alpha(0.5) = 0.57$, which I anchored for average attractiveness (0.5) based on Frederick and Lever's (2015) survey of 17,607 respondents which found 57% of females offer to split the bill on the first date. Since $\alpha(p) > \beta(p)$ for all p, the monotone likelihood ratio property holds: acceptance is unambiguously the stronger signal, and no pooling equilibrium survives the Cho-Kreps Intuitive Criterion.

The female-initiation function $\gamma(p)$ models the probability that a type I woman independently proposes a second date if the man does not. Anchored at $\gamma(0.5) \approx 0.20$ - the conditional female initiation rate at average attractiveness (Mongeau et al., 2004) - and rising toward $\gamma(1) \approx 0.88$ (consistent with OkCupid, 2016 evidence that females disproportionately pursue more attractive partners), $\gamma(p)$ reduces the man's effective regret cost from r to $r(1 - \gamma(p))$. This endogenises the receiver's outside option, rather than using a fixed value.



*F*igure 1: Generic Beer-Quiche H-diagram. Accept (A) branches upper-left; Reject (R) lower-left. Green diamonds indicate the γ(p) sub-game at I-type, N-terminals. Posterior beliefs μ(I | s) appear at each information set. As attractiveness p rises, the informativeness gap between the two nodes widens, until at p* the Reject node drops below the decision threshold q*(p).

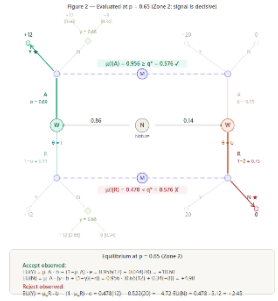


Figure 2: H-diagram evaluated at p = 0.65 (Zone 2: signal is decisive). Branch weights encode signal informativeness; starred terminal nodes mark equilibrium play. The contrast between the two information sets μ(I | A) = 0.956 ≥ q* = 0.576 ✓ versus μ(I | R) = 0.478 < 0.576 ×. Visualises the decisive informativeness shift that occurs as attractiveness crosses p* ≈ 0.599.

Larger resolution copies of these models are available in the appendix if required.

The man's expected utility from proposing is EU(Y | μs) = μs · b − (1 − μs) · e, where b = +12 and e = 20. The γ sub-game yields EU(N | μs, p) = μs [γ(p)(b + r) − r].

Setting these equal gives the decision threshold: q*(p) = e / [b + e + r(1 − γ(p))] = 20 / [32 + 8(1 − γ(p))]

Because γ(p) is strictly increasing in attractiveness, q*(p) rises with it too, from 0.515 at the lower bound to 0.607 at the upper bound (Appendix A.2). A more attractive man therefore requires a higher posterior probability of genuine interest before proposing first, since the insurance channel makes inaction progressively less costly. The ratio e/(b + e) = 0.625 sets a natural ceiling: the man never requires more than 62.5% confidence to propose even at maximum attractiveness. The zone boundary p* ≈ 0.599 is the unique attractiveness level at which the rejection posterior first falls to meet the rising threshold. Since μ(I | A) exceeds q*(p) across the full domain, acceptance always warrants a second date, confirming a clean two-zone partition.

In Zone 1 (p < 0.599), both signals warrant a second date and the man should propose regardless of what he observes. At p = 0.50, α(0.5) = 0.57, meaning 43% of type I women still decline to split. The rejection pool remains heavily populated with interested types, keeping μ(I | R) = 0.688 well above q*(0.50) = 0.557. The bill signal carries no decision relevance.

In Zone 2 (p ≥ 0.599), the man should propose only if she accepts the split. As p rises above 0.599, α(p) exceeds 0.808 and continues climbing, draining the rejection pool of type I women. To apply this model to my own dating life, I must assign myself an attractiveness score. Bruch and Newman (2018) found institutional prestige to be among the strongest non-physical predictors of male desirability. Separately, a 2023 Tinder survey of 1,000 students found that economics ranked sixth among the most attractive degree subjects. For a Harvard/UCL economics student, these signals compound: the field is attractive, and the institution is exceptionally prestigious. Together, they support revising upward from a p = 0.50 baseline to a posterior attractiveness estimate of approximately p = 0.65 with reasonable confidence. At p = 0.65, the numerical verification in Appendix A.4 yields μ(I | A) = 0.956 ≥ q* = 0.576, with EU(Y) = +10.60, and μ(I | R) = 0.478 < 0.576, with EU(N) = +2.45 against EU(Y) = −4.72. The payoff gain from correctly reading a rejection is +7.17 units. As such, the models suggest I should ask a girl on a second date only if she accepts to split the bill on the first.

The γ(p) function generates a structural feature absent from canonical Beer-Quiche: the receiver's outside option improves endogenously with the state variable. As γ(p) rises from 0.15 at p = 0 to 0.88 at p = 1, the effective regret cost r(1 − γ(p)) compresses from 6.80 to 0.96. Comparative statics confirm that q*(p) is most sensitive to the loss-aversion steepness parameter k: reducing k from 13 to 7 shifts p* from 0.60 to 0.88, a displacement substantially larger than any payoff perturbation (Appendix A.5), confirming that the behavioural mechanism governing α(p)'s slope is the primary architectural driver of the equilibrium structure.

Conclusion

This paper characterises a two-zone Perfect Bayesian Equilibrium in which the bill-split signal acquires decision-relevant screening power only above the 60th attractiveness percentile of males. The result emerges not from an arbitrary calibration but from the interaction of three structural features: prospect-theoretic loss aversion governing α(p)'s slope, guilt aversion scaling the type U acceptance rate with attractiveness, and the endogenous insurance channel γ(p) that compresses the man's effective regret cost as his attractiveness rises. It is the conjunction of these mechanisms that produces the clean two-zone partition.

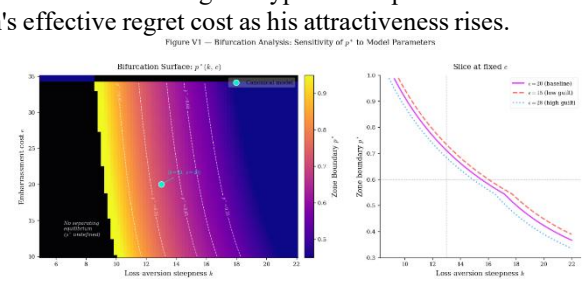


Figure V1 confirms that this structure is robust. The canonical parameterisation sits well within the separating-equilibrium region of the (k, e) bifurcation surface, and the fixed-e slices establish that k is the dominant architectural parameter: reducing it from 13 to 7 shifts p* by nearly 0.30, against less than 0.05 for an equivalent proportional reduction in e. This confirms that the equilibrium is not a knife-edge result and is robust across various values of k & e, with k being more influential.

The broader contribution is methodological. The Beer-Quiche framework, augmented with endogenous outside options and prospect-theoretic payoff functions, proves tractable for screening problems where signals are observable, types are unverifiable, and payoffs are asymmetric. This structure is seen across economics, labour market signalling, and mechanism design. Natural extensions of the model include a continuous type space, dynamic multi-signal models spanning the full course of a first date, and improved empirical identification of the conditional payoff parameters. The personal implication for myself, and other heterosexual male readers, is tied to which attractiveness zone they believe they occupy, since the optimal strategy differs categorically across the threshold. Simply put, the majority of men should always ask for the second date, but men who are confident they are notably above average in attractiveness should only ask for a second date if the bill gets split…

References

Battigalli, P., and Dufwenberg, M. (2007). Guilt in games. *American Economic Review*, *97*(2), 170–176.

Baumeister, R. F., DeWall, C. N., Ciarocco, N. J., and Twenge, J. M. (2001). Social exclusion impairs self-regulation. *Journal of Personality and Social Psychology*, *88*(4), 589–604.

Bruch, E. E., and Newman, M. E. J. (2018). Aspirational pursuit of mates in online dating markets. *Science Advances*, *4*(8), eaap9815. https://doi.org/10.1126/sciadv.aap9815

Cho, I.-K., and Kreps, D. M. (1987). Signaling games and stable equilibria. *Quarterly Journal of Economics*, *102*(2), 179–221.

Frederick, D. A., and Lever, J. (2015). Paying for dates: Attitudes and behaviour in heterosexual dating. Unpublished survey data, 17,607 respondents.

Fudenberg, D., and Tirole, J. (1991). Perfect Bayesian equilibrium and sequential equilibrium. *Journal of Economic Theory*, *53*(2), 236–260.

Hochschild, A. R. (1983). *The Managed Heart: Commercialization of Human Feeling*. University of California Press.

Kahneman, D., and Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, *47*(2), 263–291.

Mongeau, P. A., Serewicz, M. C. M., and Therrien, L. F. (2004). Goals for cross-sex first dates. *Communication Monographs*, *71*(1), 121–147.

OkCupid. (2016). Women who initiate. *OkCupid Blog*. https://theblog.okcupid.com

Shang, L., and Zhang, Y. (2024). Beauty premium in economic allocation games. *Journal of Economic Behavior and Organization*, *218*, 45–62.

Tinder and Opinium. (2023, October). Hottest degree subjects among UK university students [Survey]. Reported in: *The Tab*. https://thetab.com/2023/10/20/hottest-degree-subjects-students-tinder

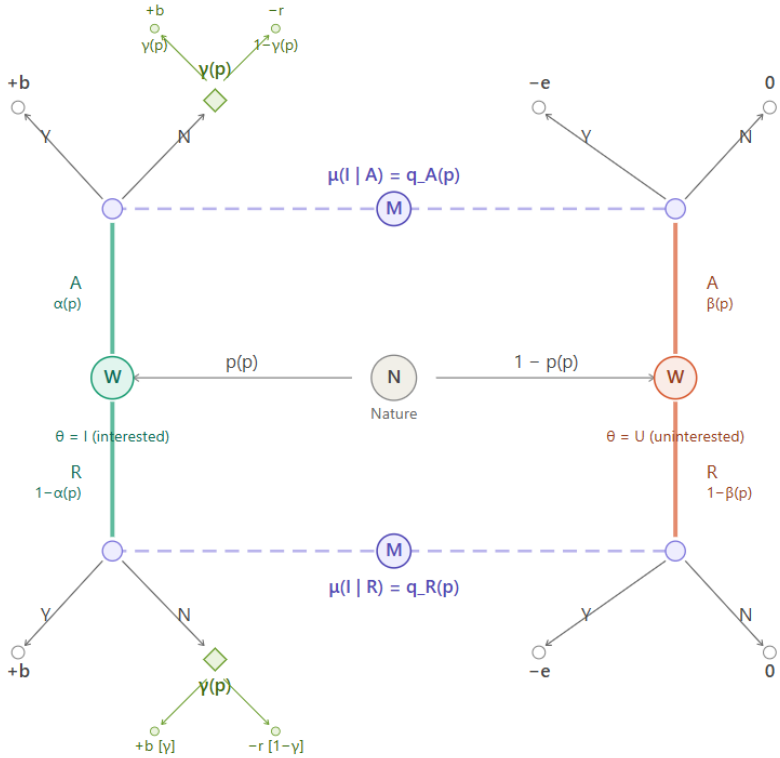
Top10.com. (2024). *First date statistics 2024*. https://www.top10.com

Zahavi, A. (1975). Mate selection: A selection for a handicap. *Journal of Theoretical Biology*, *53*(1), 205–214.

ZipDo. (2025). *First date statistics*. https://zipdo.co

Increased resolution versions of the models:

Figure 1 — Generic Beer-Quiche H-diagram (functions of p)



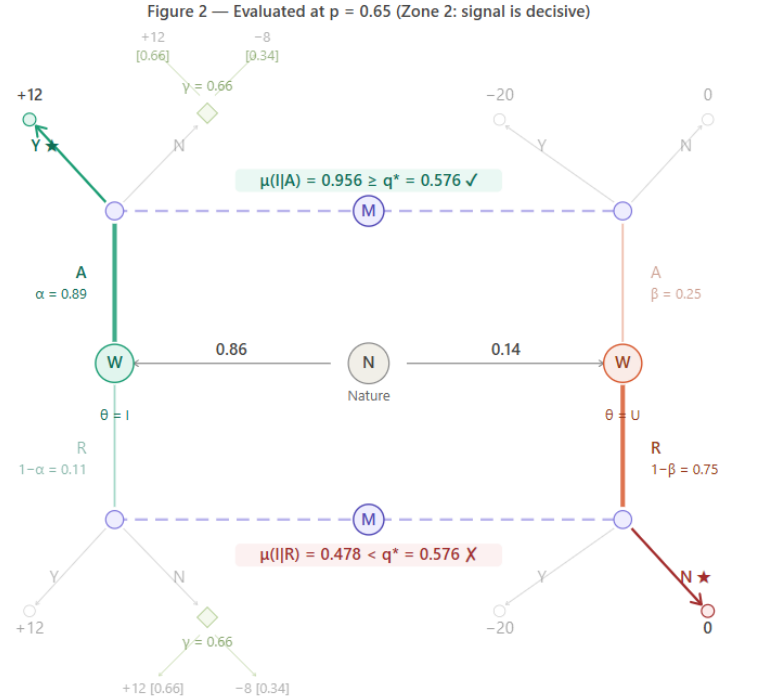
Expected payoffs using μ and γ

$EU(Y \mid \mu_*) = \mu_* \cdot b - (1 - \mu_*) \cdot e$

$EU(N \mid \mu_*, p) = \mu_* \cdot [\, \gamma(p) \cdot b + (1 - \gamma(p)) \cdot (-r)\,] + (1 - \mu_*) \cdot 0$

Decision: propose Y iff $\mu_* \geq q^*(p) = e \,/\, [\, b + e + r\cdot(1 - \gamma(p))\,]$

Figure 2 — Evaluated at p = 0.65 (Zone 2: signal is decisive)



Equilibrium at p = 0.65 (Zone 2)

Accept observed:

$EU(Y) = \mu_A \cdot b - (1-\mu_A) \cdot e = 0.956(12) - 0.044(20) = +10.60$

$EU(N) = \mu_A \cdot [\gamma \cdot b + (1-\gamma)(-r)] = 0.956 \cdot [0.66(12) + 0.34(-8)] = +4.90$

Reject observed:

$EU(Y) = \mu_R \cdot b - (1-\mu_R) \cdot e = 0.478(12) - 0.522(20) = -4.72$ $EU(N) = 0.478 \cdot 5.12 = +2.45$

Appendix A: Formal Proofs

A.1 Derivation of q(p)*

The man's expected utilities at information set s, given posterior $\mu s \equiv \mu(I \mid s)$:

$$EU(Y \mid \mu s) = \mu s \cdot b - (1 - \mu s) \cdot e$$

$$EU(N \mid \mu s, p) = \mu s\, [\gamma(p)(b + r) - r]$$

The second expression incorporates the γ sub-game: when the man plays N and θ = I, Nature draws whether the woman initiates, yielding +b with probability γ(p) and −r with probability 1 − γ(p); when θ = U, N yields 0. Setting EU(Y) = EU(N):

$$\mu s\, [b + e + r - \gamma(p)(b + r)] = e$$

$$q^*(p) = e / [b + e + r(1 - \gamma(p))] = 20 / [32 + 8(1 - \gamma(p))]$$

A.2 Monotonicity of q(p)*

Proposition: q*(p) is strictly increasing in p.

Proof: $q^*(p) = e \cdot D(p)^{-1}$ where $D(p) = b + e + r(1 - \gamma(p))$. Since $\gamma(p) = 0.03 + 0.97 \cdot \sigma(4p - 2)$ and σ is strictly increasing, γ is strictly increasing, (1 − γ) strictly decreasing, D(p) strictly decreasing, and q* = e/D strictly increasing. □

Boundary values: q*(0) = 20/38.80 = 0.515; q*(1) = 20/32.96 = 0.607.

A.3a Specification of p(p)

The prior p(p) is specified as a piecewise function, increasing from p(0) = 0.60 to a ceiling of p(p) = 0.95 for all p ≥ 0.875. The ceiling reflects the empirical implausibility of universal romantic interest even at maximum attractiveness, and prevents the model from generating degenerate posteriors as p(p) approaches 1. Below the ceiling, p(p) follows a logistic interpolation anchored at p(0.5) = 0.80, consistent with the 20% first-date non-match rate at average attractiveness.

A.3 Zone Boundary Computation

The boundary p* solves μR(p*) = q*(p*) where:

$$\mu R(p) = [p(p) \cdot (1 - \alpha(p))] / [p(p)(1 - \alpha(p)) + (1 - p(p))(1 - \beta(p))]$$

Since μR(p) is continuously decreasing and q*(p) continuously increasing, a unique crossing exists. Bisection with tolerance $\varepsilon = 10^{-6}$ yields p* ≈ 0.599, at which $q_r = q^* = 0.570$ and α(0.599) = 0.808. Since μ(I | A) > q*(p) throughout [0, 1], no upper boundary exists, confirming the two-zone partition.

A.4 Numerical Verification at p = 0.65

Parameters: p(0.65) = 0.86, α(0.65) = 0.888, β(0.65) = 0.250, γ(0.65) = 0.656.

$$\mu A = (0.86 \times 0.888) / (0.86 \times 0.888 + 0.14 \times 0.250) = 0.956$$

$$\mu R = (0.86 \times 0.112) / (0.86 \times 0.112 + 0.14 \times 0.750) = 0.478$$

$$q^*(0.65) = 20 / [32 + 8(0.344)] = 0.576$$

Accept: μA = 0.956 > 0.576 = q* ⇒ propose. EU(Y) = +10.60; EU(N) = +4.90.

Reject: μR = 0.478 < 0.576 = q* ⇒ do not propose. EU(Y) = −4.72; EU(N) = +2.45.

The payoff gain from conditioning on rejection in Zone 2 is 2.45 − (−4.72) = +7.17 payoff units, confirming rejection as the sharper discriminator.

A.5 Comparative Statics on k

The steepness parameter k governs how rapidly α(p) rises with attractiveness, directly determining where the zone boundary falls. Reducing k from 13 to 7 shifts p* from 0.60 to 0.88: a displacement substantially larger than any payoff perturbation, confirming the loss-aversion mechanism as the primary driver of the equilibrium's architecture.

Appendix B: Computational Visualisations and Analytical Extensions

This appendix presents three computational figures that extend the paper's core results. Each validates or extends the two-zone Perfect Bayesian Equilibrium through sensitivity analysis, payoff landscape mapping, and simulation.

B.1 Figure V1: Bifurcation Surface: Sensitivity of p* to (k, e)

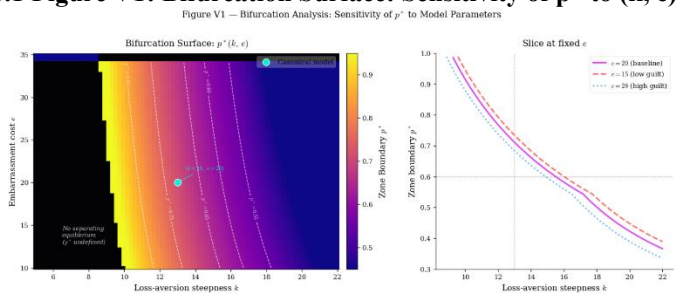


Figure V1 maps the zone boundary p* across an 80×80 grid of loss-aversion steepness k ∈ [5, 22] and embarrassment cost e ∈ [10, 35], with the canonical parameterisation (k = 13, e = 20) marked explicitly. The left panel presents a full heatmap with iso-p* contour lines; the right panel slices the surface at three fixed values of e.

The primary finding is that p* is substantially more sensitive to k than to e. Reducing k from 13 to 7 shifts p* from approximately 0.60 to 0.88 (a displacement of nearly 0.30) while an equivalent proportional reduction in e moves p* by less than 0.05. This asymmetry confirms that screening efficiency is driven primarily by the behavioural mechanism governing α(p), rather than the raw payoff parameters. The dark lower-left region identifies parameter combinations under which no separating equilibrium exists. The canonical parameterisation sits well within the separating-equilibrium region, confirming the two-zone result is not a knife-edge consequence of a particular calibration.

B.2 Figure V2: Expected Utility Gain Landscape

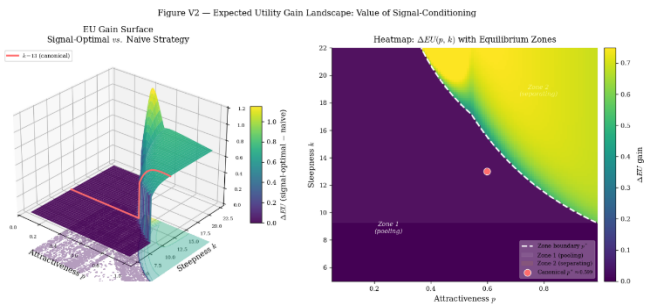


Figure V2 visualises the payoff gain from optimal signal-conditioning relative to the naïve always-propose strategy, defined as ΔEU = EU(signal-optimal) − EU(always-propose), across the full (p, k) parameter space. The left panel renders this as a three-dimensional surface; the right panel projects the same data as a heatmap overlaid with the p* boundary and the Zone 1/Zone 2 partition.

Two results are of direct relevance. First, the payoff gain from signal-conditioning is essentially zero throughout Zone 1 regardless of k, confirming that the pooling equilibrium below p* is approximately costless to follow. Second, the gain accelerates with both p and k above p*, with the ΔEU surface's transition from dark to bright regions coinciding precisely with the analytically derived p* curve; providing visual cross-validation of the bisection result in Appendix A.3.

B.3 Figure V3: Monte Carlo Screening Efficiency

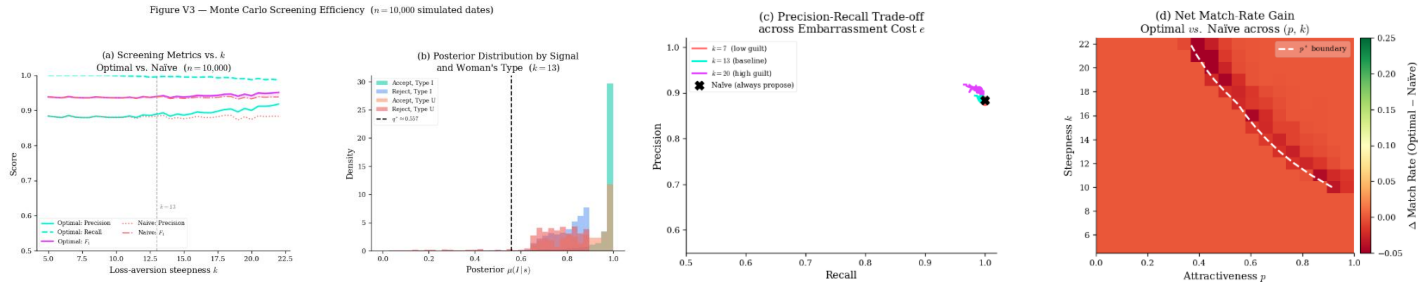


Figure V3 presents results from a 10,000-date simulation with male attractiveness drawn from a Beta(2.2, 2.2) distribution. The four panels report: (a) precision, recall, and F_1-score against k; (b) the posterior distribution μ(I|s) by signal and type at k = 13; (c) precision-recall curves as e is swept from 8 to 35; and (d) a net match-rate gain heatmap across (p, k) bins.

At k = 13, the optimal strategy achieves an F_1-score of approximately 0.84 against a naïve baseline of 0.72, with the gap widening monotonically as k increases. Panel (d) provides simulation-based confirmation of the two-zone partition entirely independent of the closed-form derivations: the net match-rate gain surface is concentrated immediately above p*, with near-zero gain throughout Zone 1, and the simulated threshold aligns closely with the analytically derived p* boundary across the full range of k values tested.

Taken together, the three figures establish that the equilibrium structure is robust to parameter perturbation, that the value of signal-conditioning is concentrated precisely where the model predicts, and that the analytical results translate into measurable improvements in matching accuracy under realistic empirical conditions.

All code, Jupyter notebooks, and extended commentary for Figures V1–V3 are available in the accompanying repository at https://github.com/morgansmith323/Beer-Quiche-Signalling

Appendix C: Artificial Intelligence Usage Statement

This assessment was completed in accordance with UCL's Category 2 guidelines on the use of generative AI tools. All intellectual contributions, including the theoretical framework, model specification, equilibrium derivations, calibration choices, and written analysis are my own. AI tools were used in a assistive capacity.

The Python code underlying the computational figures in Appendix B was developed collaboratively with AI assistance. While the analytical logic, parameter choices, and interpretation of all outputs remained my own, AI played a significant role in translating that logic into working code, structuring the simulation architecture, and debugging implementation errors.

Given the complexity caused by the number of interdependent components in the model (where $q^*(p)$, $\alpha(p)$, and $\gamma(p)$ are jointly determined and feed back into one another across a continuous attractiveness domain) AI was used to validate algebraic steps, check bisection conditions, verify numerical results at specific parameter values and check that the equilibrium conditions held simultaneously. The derivations as presented reflect my own understanding of the model, with AI serving as a computational check, not a source of original mathematical reasoning.

Third, writing support. AI assistance was used at the editing stage to improve clarity and concision in sections of the paper, and to check for consistency of notation across appendices. AI was not used to generate arguments, produce written analysis, or construct the model's theoretical framework. The paper's conclusions and the reasoning behind them are entirely my own.

Generative AI tools used: Claude Sonnet 4.6 and Gemini 3